Srinidh Ameerpeta

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SUMMARY

3rd-year Integrated M.Tech CS student at University of Hyderabad with hands-on experience building and deploying production-grade ML systems end-to-end. Designed and shipped a live multi-modal anomaly detection platform (GNN + SAC RL + LSTM) on real-time Kafka streams, served via FastAPI, with an interactive Streamlit dashboard. Strong foundation in distributed systems, data structures, and scalable backend architecture. Comfortable across the full stack from model training to containerised deployment.

EDUCATION

University of Hyderabad 2023 - 2028

Integrated M.Tech in Computer Science CGPA: 8.07 Hyderabad, Telangana

- Relevant coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Web Technologies.
- AI and ML Training Certification - IIIT Hyderabad.

Rajiv Gandhi University of Knowledge Technologies, Basar 2021 - 2023

Pre-University (Class XI - XII) CGPA: 9.25 Basar, Telangana

TECHNICAL SKILLS

ML / AI: PyTorch, PyTorch Geometric (GNN, GraphSAGE, GAT), Stable-Baselines3 (SAC RL), Scikit-learn, LangChain, FAISS, Qdrant, YOLOv8, LSTM, Isolation Forest

Languages: Python, C++, Java, JavaScript, SQL, C

Backend: FastAPI, Node.js, Express.js, Apache Kafka, REST APIs, Microservices, Docker

Databases: MongoDB, MySQL, Neo4j, InfluxDB

Frontend: React, Vite, Chart.js, Streamlit

Systems: Linux, Git, Docker Compose, Nginx, Postman, LLM Prompt Engineering

RESEARCH & TECHNICAL PROJECTS

HydroTwin OS - Industrial Digital Twin with Multi-Modal Anomaly Detection *PyTorch · Stable-Baselines3 · PyG · Neo4j · LangChain · Kafka · FastAPI · Streamlit*

[Live Demo] [GitHub]

- Designed a physics-informed digital twin of water and cooling infrastructure for data centres; enforced thermodynamic constraints by incorporating heat-transfer PDE residuals directly into the neural network loss (PINN formulation), ensuring physical conservation laws hold under sensor noise.
- Built a multi-modal anomaly detection pipeline fusing three sensor streams (thermal IR video via YOLOv8 at 30 fps, vibration/motor current via 1D CNN at 1 kHz, and chemical sensors via LSTM at 1 Hz) through a late-fusion transformer; localises pump failures and coolant leaks in real time.
- Implemented a Soft Actor-Critic (SAC) reinforcement learning agent using Stable-Baselines3 for continuous cooling setpoint optimisation, balancing Water Usage Effectiveness, Power Usage Effectiveness, and carbon intensity under hard thermal safety constraints.
- Modelled the infrastructure as a property graph in Neo4j and trained a PyTorch Geometric GNN (GraphSAGE + GAT layers) for spatial fault localisation; faults propagate across graph edges, enabling cascade detection that single-sensor approaches miss.
- Engineered a LangChain RAG pipeline backed by InfluxDB and a Qdrant vector store to generate operator-readable root-cause explanations from maintenance manuals and regulatory documents in response to anomaly alerts.
- Deployed the full system on Apache Kafka for real-time sensor stream ingestion; served via FastAPI; interactive monitoring dashboard built in Streamlit with live anomaly overlays.

StructOptima - Automated Structural Analysis and Design Engine *Python · NumPy · Streamlit · Plotly · ezdxf · pytest*

[Live Demo] [GitHub]

- Engineered a 51-module Python engine that ingests architectural CAD files (DXF/IFC) and generates a full design package including column placement, beam sizing, slab design, foundation detailing, and rebar schedules without manual intervention.
- Implemented a 2D finite element analysis (FEA) solver from scratch using the stiffness matrix method for frame deflection and internal force computation; integrated seismic and wind load pipelines with factored load combinations.

- Built a multi-stage optimisation pipeline for cost minimisation across column layouts, with automated safety constraint enforcement, slenderness ratio checks, and 13+ black-box risk management routines.
- Designed a layered architecture (parsing, analysis, output) with DXF/IFC import, Plotly 3D visualisation, and PDF/Excel/DXF report generation; shipped a Streamlit web app with an interactive column editor and live design feedback.
- Wrote a 28-suite pytest test harness covering unit, integration, and edge-case tests across all core calculation modules.

GreenScore - AI-Driven Sustainability Modelling Platform *React · Node.js · FastAPI · Python · Docker · MongoDB · Ollama / Llama 3*
[GitHub]

- Architected a three-service Docker Compose application (React/Vite frontend, Node.js/Express REST API, Python FastAPI AI service) with Nginx reverse proxy; JWT authentication and Joi schema validation throughout.
- Engineered an LLM extraction pipeline (Ollama-hosted Llama 3 8B) to parse ambiguous natural language activity descriptions, including Hinglish and regional food terminology, into structured Scope 1, 2, and 3 carbon emission records with confidence scores.
- Designed a structured data pipeline converting free-text inputs to validated JSON emission records; built a Chart.js analytics dashboard showing emission breakdowns, trends, and monthly goal tracking.
- Identified the domain-shift failure mode where extraction accuracy degrades on out-of-distribution inputs (code-mixed text, regional terminology), a robustness problem connecting directly to cross-domain NLP adaptation research.

Greenwashing Detector - NLP-Based Corporate Claim Classifier *Python · Streamlit · NLP*
[GitHub]

- Built a text classification system to detect misleading environmental claims in corporate sustainability reports; deployed as an interactive Streamlit web application.
- Addresses the weakly supervised labelling challenge for implicit greenwashing language, specifically vague claims that are technically true but contextually misleading.

UoH Alumni Network Scraper - Large-Scale Dynamic Web Extraction *Node.js · Express.js · Puppeteer · Server-Sent Events*
[GitHub]

- Reverse-engineered a dynamically rendered AngularJS alumni portal and implemented headless-browser extraction via Puppeteer to build a structured alumni dataset, bypassing AJAX pagination and lazy-loaded content.
- Implemented Server-Sent Events (SSE) for real-time streaming of scraping progress to a client dashboard; structured output as analytically queryable JSON/CSV.

INDEPENDENT STUDY & RESEARCH READING

- Regularly study distributed systems concepts: consensus algorithms, fault tolerance, replication strategies, and consistency models; working through foundational texts including Designing Data-Intensive Applications (Kleppmann).
- Study cloud computing architecture: containerisation (Docker, Kubernetes concepts), microservices design patterns, event-driven systems (Kafka), and cloud-native infrastructure trade-offs.
- Active algorithmic problem-solving on LeetCode; focus on advanced data structures, graph algorithms, dynamic programming, and system design problems.

ACHIEVEMENTS & ACTIVITIES

- AI and ML Training Certification - IIIT Hyderabad.
- Built and publicly deployed two live ML/software systems independently as a 3rd-year undergraduate: HydroTwin OS and StructOptima.
- Mathematics tutor (up to Class 12), focusing on analytical problem-solving and structured reasoning.
- Interests: astronomy and astrophysics; non-fiction on science, technology, and systems thinking.